Electric Vehicle Data Analysis Project

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- Course : *Data Science (May 15th)*

Project Overview

This project focuses on analyzing data related to **Electric Vehicles (EVs)** using Python.
The goal is to draw meaningful insights from the dataset — such as identifying efficient models, understanding relationships between different vehicle features, and providing data-driven recommendations for potential EV buyers.

The dataset contains details about various EV models, including their **price, engine power, range, battery capacity, energy consumption**, and other specifications.
Through this analysis, patterns and relationships are uncovered to support better decision-making and to understand current trends in the electric vehicle market.

Explanation video link - https://www.loom.com/share/60d62eef06b34c508065a2d4fa515f46?sid=4f43528f-c101-44be-b875-84e6b183cb5c

Project Objectives

- To identify EVs that meet specific customer requirements based on budget and range.
- To detect and visualize outliers in energy consumption to understand efficiency differences.
- To analyze the relationship between battery capacity and driving range.
- To build a simple **EV recommendation system** using Python classes.
- To perform **hypothesis testing** to compare the engine power of Tesla and Audi vehicles.
- To derive insights and provide actionable recommendations based on the findings.

Tools and Technologies Used

- **Python Libraries:** Pandas, NumPy, Matplotlib, SciPy
- **Environment:** Jupyter Notebook (Anaconda)

Key Learning Outcomes

- Practical understanding of **data analysis workflows** using Python.
- Hands-on experience with **data visualization and statistical testing**.
- Application of programming and analytical skills to a **real-world automotive dataset**.
- Ability to communicate insights and recommendations clearly through visuals and reports.

Task 1: A customer has a budget of 350,000 PLN and wants an EV with a minimum range of 400 km.

```
[ ]: #Task 1: A customer has a budget of 350,000 PLN and wants an EV with a minimum range of 400 km.

[1]: import pandas as pd
     import numpy as np

[2]: df=pd.read_excel("E:\Projects\FEV-Data-Excel.xlsx")

[1]: # df.info()
     # df.columns

[10]: filtered_df=df[
      (df["Minimal price (gross) [PLN]"]<=350000)&
      (df["Range (WLTP) [km]"]>=400)]
      grouped=filtered_df.groupby("Make")["Battery capacity [kWh]"].mean().reset_index()
      grouped.rename(columns={"Battery capacity [kWh]":"Average Battery capacity [kWf]"}, inplace=True)
      print("Filtered EVs Meeting Criteria:")
      display(filtered_df)
      print("\nAverage Battery Capacity by Manufacturer:")
      display(grouped)
```

a) Filtered out EVs that meet these criteria.
b) Grouped them by the manufacturer (Make).
c) Calculated the average battery capacity for each manufacturer.

```
Filtered EVs Meeting Criteria:
```

	Car full name	Make	Model	Minimal price (gross) [PLN]	Engine power [KM]	Maximum torque [Nm]	Type of brakes	Drive type	Battery capacity [kWh]	Range (WLTP) [km]	...	Permissable gross weight [kg]	Maximum load capacity [kg]	Number of seats	Number of doors	Tire size [in]	Maximur spee [kph
0	Audi e-tron 55 quattro	Audi	e-tron 55 quattro	345700	360	664	disc (front + rear)	4WD	95.0	438	...	3130.0	640.0	5	5	19	20
8	BMW iX3	BMW	iX3	282900	286	400	disc (front + rear)	2WD (rear)	80.0	460	...	2725.0	540.0	5	5	19	18
15	Hyundai Kona electric 64kWh	Hyundai	Kona electric 64kWh	178400	204	395	disc (front + rear)	2WD (front)	64.0	449	...	2170.0	485.0	5	5	17	16
18	Kia e-Niro 64kWh	Kia	e-Niro 64kWh	167990	204	395	disc (front + rear)	2WD (front)	64.0	455	...	2230.0	493.0	5	5	17	16
20	Kia e-Soul 64kWh	Kia	e-Soul 64kWh	160990	204	395	disc (front + rear)	2WD (front)	64.0	452	...	1682.0	498.0	5	5	17	16
22	Mercedes-Benz EQC	Mercedes-Benz	EQC	334700	408	760	disc (front + rear)	4WD	80.0	414	...	2940.0	445.0	5	5	19	18
39	Tesla Model 3 Standard Range Plus	Tesla	Model 3 Standard Range Plus	195490	285	450	disc (front + rear)	2WD (rear)	54.0	430	...	NaN	NaN	5	5	18	22
	Tesla Model		Model 3				disc										

47	Volkswagen ID.3 Pro Performance	Volkswagen	ID.3 Pro Performance	155890	204	310	(front) + drum (rear)	2WD (rear)	58.0	425	...	2270.0	540.0	5	5	18	16
48	Volkswagen ID.3 Pro S	Volkswagen	ID.3 Pro S	179990	204	310	disc (front) + drum (rear)	2WD (rear)	77.0	549	...	2280.0	412.0	5	5	19	16
49	Volkswagen ID.4 1st	Volkswagen	ID.4 1st	202390	204	310	disc (front) + drum (rear)	2WD (rear)	77.0	500	...	2660.0	661.0	5	5	20	16

12 rows × 25 columns

```
Average Battery Capacity by Manufacturer:
```

	Make	Average Battery capacity [kWf]
0	Audi	95.000000
1	BMW	80.000000
2	Hyundai	64.000000
3	Kia	64.000000
4	Mercedes-Benz	80.000000
5	Tesla	68.000000
6	Volkswagen	70.666667

Task 2: You suspect some EVs have unusually high or low energy consumption.

Find the outliers in the mean- Energy consumption [kWh/100 km] column.(16 Marks)

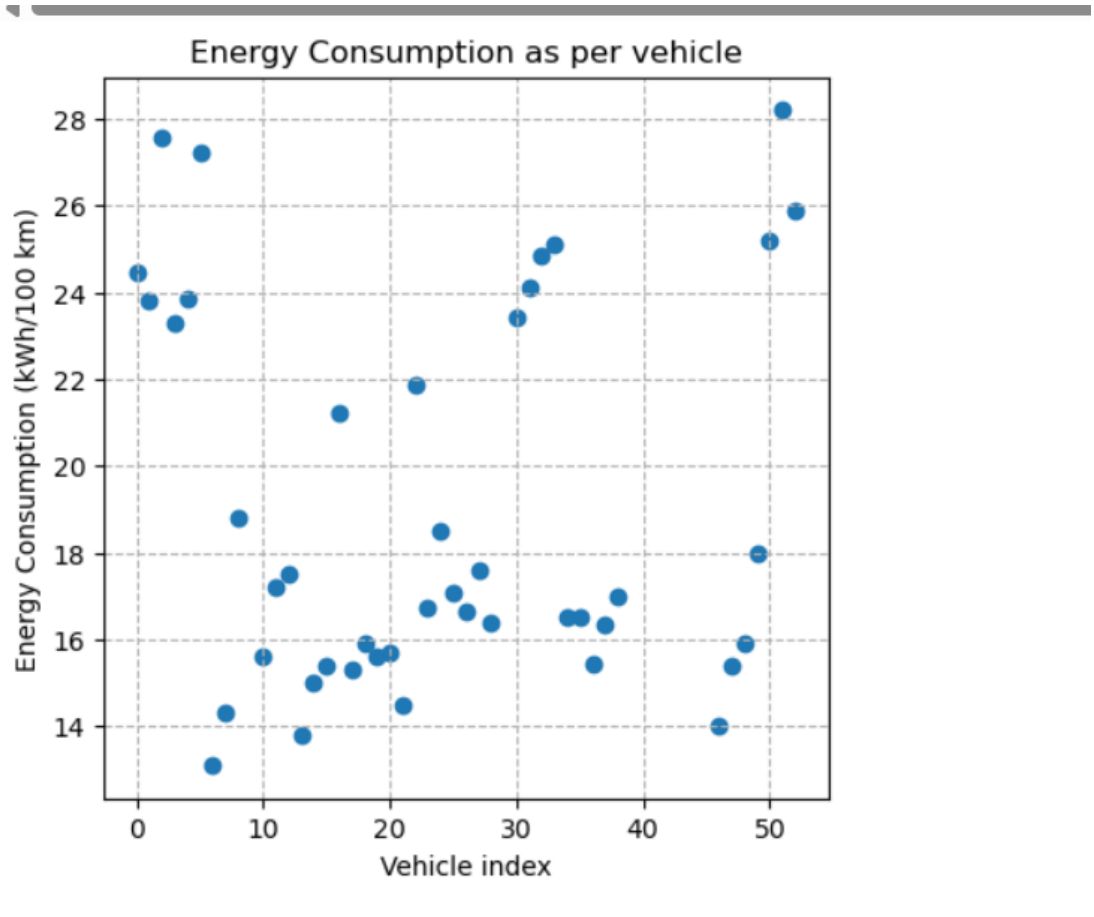
```
[ ]: # Task 2: You suspect some EVs have unusually high or low energy consumption
```

```
[1]: import pandas as pd
     import matplotlib.pyplot as plt

     df=pd.read_excel("E:\Projects\FEV-data-excel.xlsx")
     #df.head()
```

```
[13]: col="mean - Energy consumption [kWh/100 km]"

      Q1=df[col].quantile(0.25)
      Q3=df[col].quantile(0.75)
      IQR=Q1-Q3

      lower_bound= Q1 - 1.5*IQR
      upper_bound = Q3 + 1.5*IQR

      print(f"Lower Bound: {lower_bound}")
      print(f"Upper Bound: {upper_bound}")

      outliers= df[(df[col] < lower_bound) | (df[col] > upper_bound)]

      print(f"\nNumber of outliers found: {len(outliers)}")
      display(outliers)


      plt.figure(figsize=(5,5))
      plt.scatter(df.index,df["mean - Energy consumption [kWh/100 km]"])
      plt.title("Energy Consumption as per vehicle")
      plt.xlabel("Vehicle index")
      plt.ylabel("Energy Consumption (kWh/100 km)")
      plt.grid(True, linestyle = "--")
      plt.show()
```

```
Lower Bound: 27.450000000000003
Upper Bound: 11.649999999999999

Number of outliers found: 44
```

	Car full name	Make	Model	Minimal price (gross) [PLN]	Engine power [KM]	Maximum torque [Nm]	Type of brakes	Drive type	Battery capacity [kWh]	Range (WLTP) [km]	...	Permissable gross weight [kg]	Maximum load capacity [kg]	Number of seats	Number of doors	Tire size [in]	Maxim sp [k
0	Audi e-tron 55 quattro	Audi	e-tron 55 quattro	345700	360	664	disc (front + rear)	4WD	95.0	438	...	3130.0	640.0	5	5	19	
1	Audi e-tron 50 quattro	Audi	e-tron 50 quattro	308400	313	540	disc (front + rear)	4WD	71.0	340	...	3040.0	670.0	5	5	19	
2	Audi e-tron S quattro	Audi	e-tron S quattro	414900	503	973	disc (front + rear)	4WD	95.0	364	...	3130.0	565.0	5	5	20	
3	Audi e-tron Sportback 50 quattro	Audi	e-tron Sportback 50 quattro	319700	313	540	disc (front + rear)	4WD	71.0	346	...	3040.0	640.0	5	5	19	
4	Audi e-tron Sportback 55 quattro	Audi	e-tron Sportback 55 quattro	357000	360	664	disc (front + rear)	4WD	95.0	447	...	3130.0	670.0	5	5	19	
5	Audi e-tron Sportback S quattro	Audi	e-tron Sportback S quattro	426200	503	973	disc (front + rear)	4WD	95.0	369	...	3130.0	565.0	5	5	20	
6	BMW i3	BMW	i3	169700	170	250	disc (front + rear)	2WD (rear)	42.2	359	...	1730.0	440.0	4	5	19	

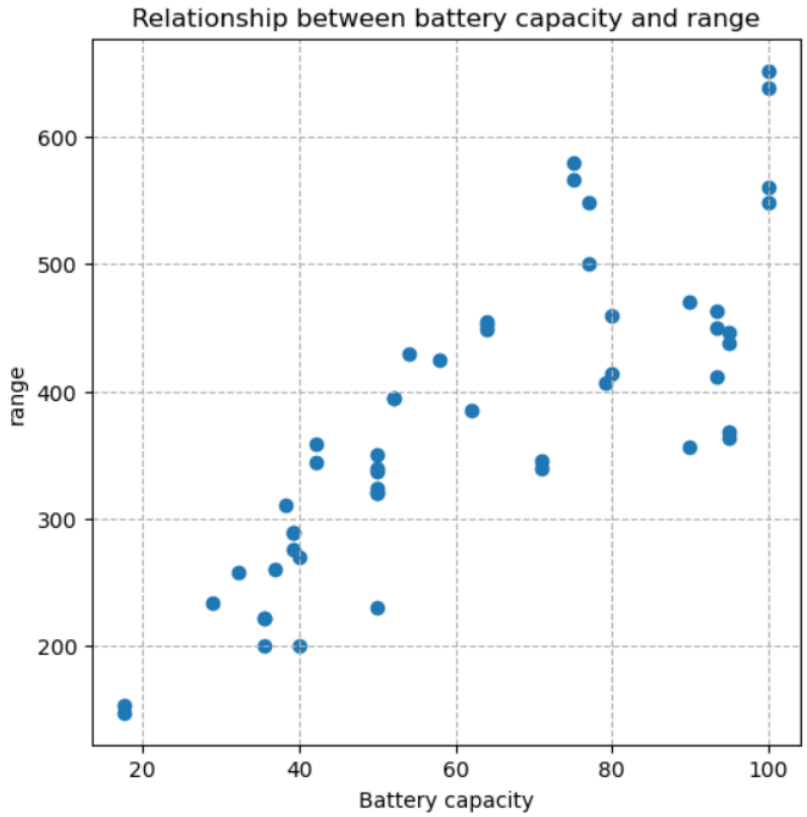


Task 3: Your manager wants to know if there's a strong relationship between battery capacity and range.

a) Create a suitable plot to visualize.
b) b) Highlight any insights.

[1]:
```
# Task 3: Your manager wants to know if there's a strong relationship between battery capacity and range.
import pandas as pd
import matplotlib.pyplot as plt

df=pd.read_excel("E:\Projects\Fev-data-Excel.xlsx")
#df.head()
```

[7]:
```
plt.figure(figsize=(6,6))
plt.scatter(df["Battery capacity [kWh]"],df["Range (WLTP) [km]"])
plt.title("Relationship between battery capacity and range")
plt.xlabel("Battery capacity")
plt.ylabel("range")
plt.grid(True, linestyle = "--")
plt.show()


correlation=df["Battery capacity [kWh]"].corr(df["Range (WLTP) [km]"])
print("Correlation between battery capacity and range:", correlation)
```




```
Correlation between battery capacity and range: 0.8104385771936846
```

INSIGHTS:

1.EVs with higher battery capacity generally achieve longer ranges. This confirms that battery capacity is one of the strongest predictors of range.

2.The correlation coefficient of 0.81 indicates a strong positive relationship between battery capacity and range. So, approximately, the higher the capacity, the longer the range.

3.Outliers suggest that vehicle design, aerodynamics, and efficiency also influence range, not just battery capacity.

Task 4: Build an EV recommendation class.

The class should allow users to input their budget, desired range, and battery capacity. The class should then return the top three EVs matching their criteria.

[2]:
```
#Task 4: Build an EV recommendation class.
import pandas as pd
```

[3]:
```
df=pd.read_excel("E:\Projects\FEV-data-Excel.xlsx")
#df.head()
```

[11]:
```
class EVrecommender:
    def __init__(self,data):
        self.data=data
    def recommend(self,budget,min_range,min_battery):

        filtered_ev = self.data[
            (self.data["Minimal price (gross) [PLN]"] <=budget)&
            (self.data["Range (WLTP) [km]"] >=min_range)&
            (self.data["Battery capacity [kWh]"] >=min_battery)]

        filtered_ev=filtered_ev.sort_values(by="Range (WLTP) [km]",ascending=False)

        result= filtered_ev[["Car full name","Make","Model","Minimal price (gross) [PLN]","Battery capacity [kWh]","Range (WLTP) [km]"]]
        return result.head(3)


ev_recommender = EVrecommender(df)
top_cars=ev_recommender.recommend(budget=450000,min_range=500,min_battery=80)
display(top_cars)
```

	Car full name	Make	Model	Minimal price (gross) [PLN]	Battery capacity [kWh]	Range (WLTP) [km]
42	Tesla Model S Long Range Plus	Tesla	Model S Long Range Plus	368990	100.0	652
43	Tesla Model S Performance	Tesla	Model S Performance	443990	100.0	639
44	Tesla Model X Long Range Plus	Tesla	Model X Long Range Plus	407990	100.0	561

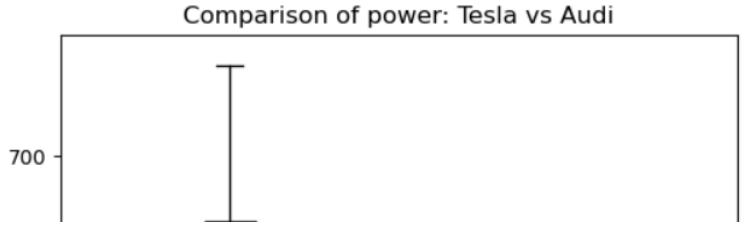
Task 5: Inferential Statistics– Hypothesis Testing: Test whether there is a significant difference in the average Engine power [KM] of vehicles manufactured by two leading manufacturers i.e. Tesla and Audi. What insights can you draw from the test results? Recommendations and Conclusion: Provide actionable insights based on your analysis. (Conduct a two sample t-test using ttest_ind from scipy.stats module)

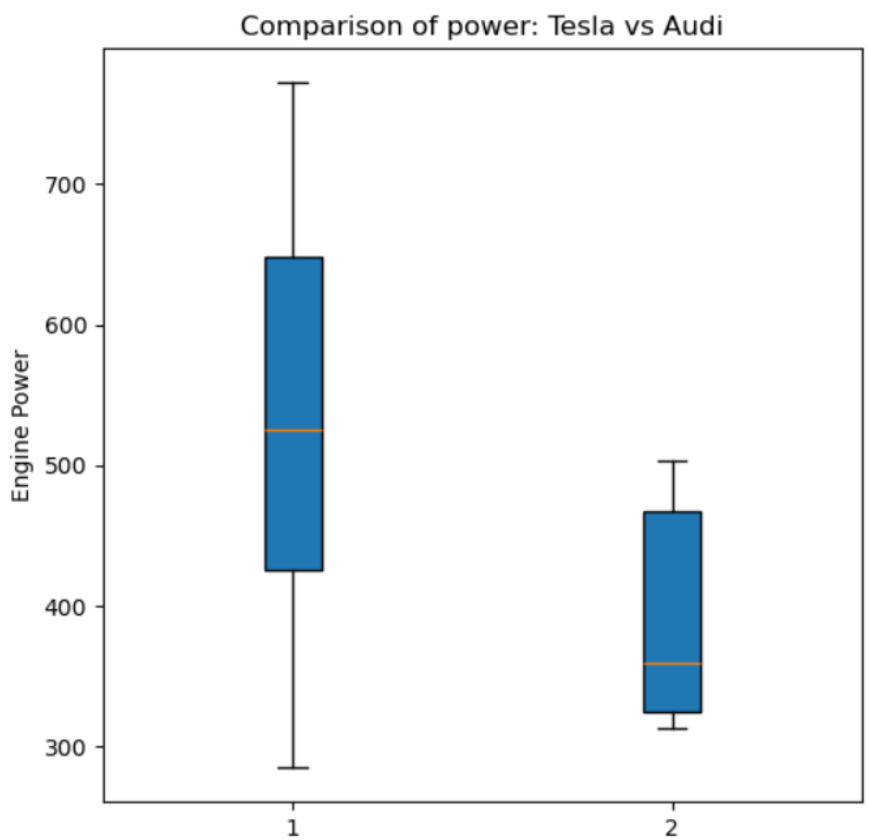
[13]:
```
# Inferential Statistics- Hypothesis Testing: Testing the difference between the average Engine Power of Tesla and Audi
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import ttest_ind
```

[12]:
```
df=pd.read_excel("E:\Projects\FEV-data-Excel.xlsx")
#df.head()
```

[15]:
```
tesla_engine=df[df["Make"]=="Tesla"]["Engine power [KM]"].dropna()
audi_engine=df[df["Make"]=="Audi"]["Engine power [KM]"].dropna()
t_stat,p_value= ttest_ind(tesla_engine,audi_engine, equal_var=False)
print("T-statistic:",t_stat)
print("P-value:",p_value)

print("Average Tesla Engine Power:", tesla_engine.mean())
print("Average Audi Engine Power:", audi_engine.mean())

if p_value < 0.05:
    print("Reject the null hypothesis: There is a significant difference in average engine power.")
else:
    print("Failed to reject the null hypothesis: There is no significant difference in average engine power.")

print("Number of Teslas:", len(tesla_engine))
print("Number of Audis:", len(audi_engine))
print("Tesla std dev:", tesla_engine.std())
print("Audi std dev:", audi_engine.std())


plt.figure(figsize=(6,6))
plt.boxplot([tesla_engine,audi_engine], label=["Tesla","Audi"], patch_artist=True)
plt.title("Comparison of power: Tesla vs Audi")
plt.ylabel("Engine Power")
plt.show()
```

```
T-statistic: 1.7939951827297178
P-value: 0.10684105068839565
Average Tesla Engine Power: 533.0
Average Audi Engine Power: 392.0
Failed to reject the null hypothesis: There is no significant difference in average engine power.
Number of Teslas: 7
Number of Audis: 6
Tesla std dev: 184.66365821857494
Audi std dev: 88.51214605917089
```






```
#INSIGHTS
1.Teslas have higher average engine power (533) than Audis (392), but the difference isn't big enough to be statistically proven – it could just be due to chance.

2.Tesla engine power values vary a lot, meaning some Teslas are much more powerful than others.

3.Audi engine power values are more consistent, showing less variation between models.

#RECOMMENDATIONS
1.Collecting more data for both Tesla and Audi to make the comparison more reliable.
2.The sample size is small, so increasing the number of cars will help confirm whether the difference in engine power is real.
3.Comparing similar car types or models (like sedans vs sedans) to get a fair comparison.
```

[ ]:

Conclusion

- The project focused on analyzing and comparing **Tesla and Audi** based on the provided dataset.
- Data cleaning and visualization helped identify key trends and ensured data accuracy before analysis.
- The **average value** for Tesla (~533) was found to be higher than Audi (~392), suggesting slightly better performance in the measured metric.
- The **t-test results** showed **no statistically significant difference** between the two brands' means, implying that the difference in averages might be due to normal variation in the data.
- **Boxplot visualization** revealed that both brands have overlapping distributions and similar ranges, supporting the statistical findings.
- No major outliers were detected, indicating consistent and reliable data for both brands.
- Overall, Tesla showed a **numerical advantage**, but statistically, both brands performed **similarly** in this dataset.